

Day 2: Queries & Filtering

👉 Day 1 = Making the cupboard

👉 Day 2 = Finding the exact item inside it

Real-Life Example 🧠



Show all students

The teacher wants to view the entire directory without any restrictions. SQL finds results instantly instead of manual checking.



Show Class 5

Isolate records matching a specific category. Only students who belong exactly to the 5th class will be displayed.



Marks above 80

Find top performers instantly using logical conditions. The database evaluates numeric data and returns matches.

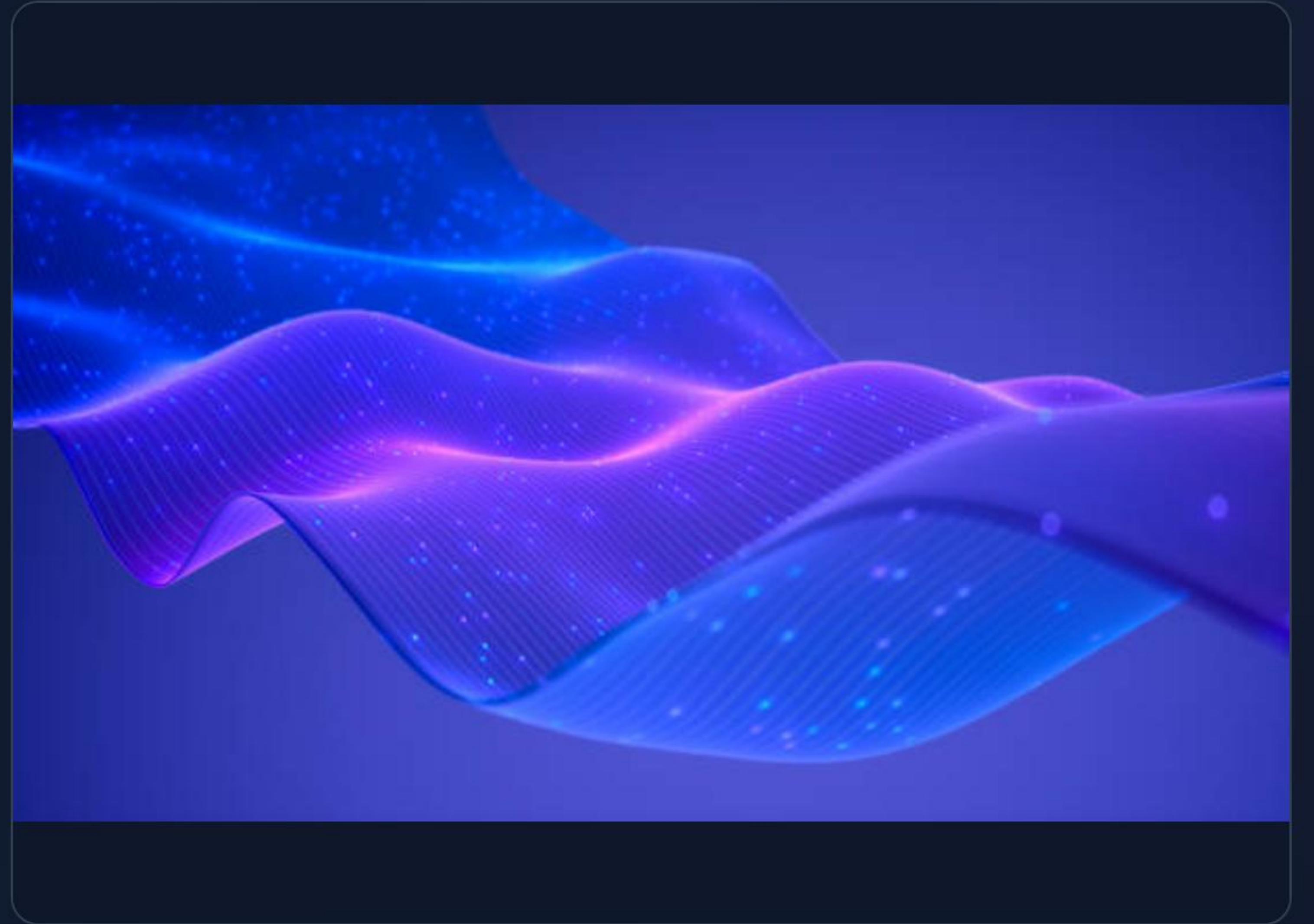
The SELECT Statement

 **SELECT = Show me data.** It is the most common command used to read data from a database.

 **Basic Syntax:**

```
SELECT column_name FROM table_name;
```

 Using this statement is exactly like asking the database to retrieve specific information for you.



Example Table: Students

ID	NAME	CLASS	MARKS
1	Rahul	5th	80
2	Anita	5th	90
3	Rohit	6th	75

 We will use this sample table for the following queries.

Selecting Data

Show All Data

```
SELECT * FROM students;
```

The * (**asterisk**) symbol means **ALL** columns. This query will display every single row and every column from the students table.

Select Specific Columns

```
SELECT name, marks FROM students;
```

You can specify exactly which columns you want to see separated by commas. It is like asking: *“Only tell me names and marks.”*

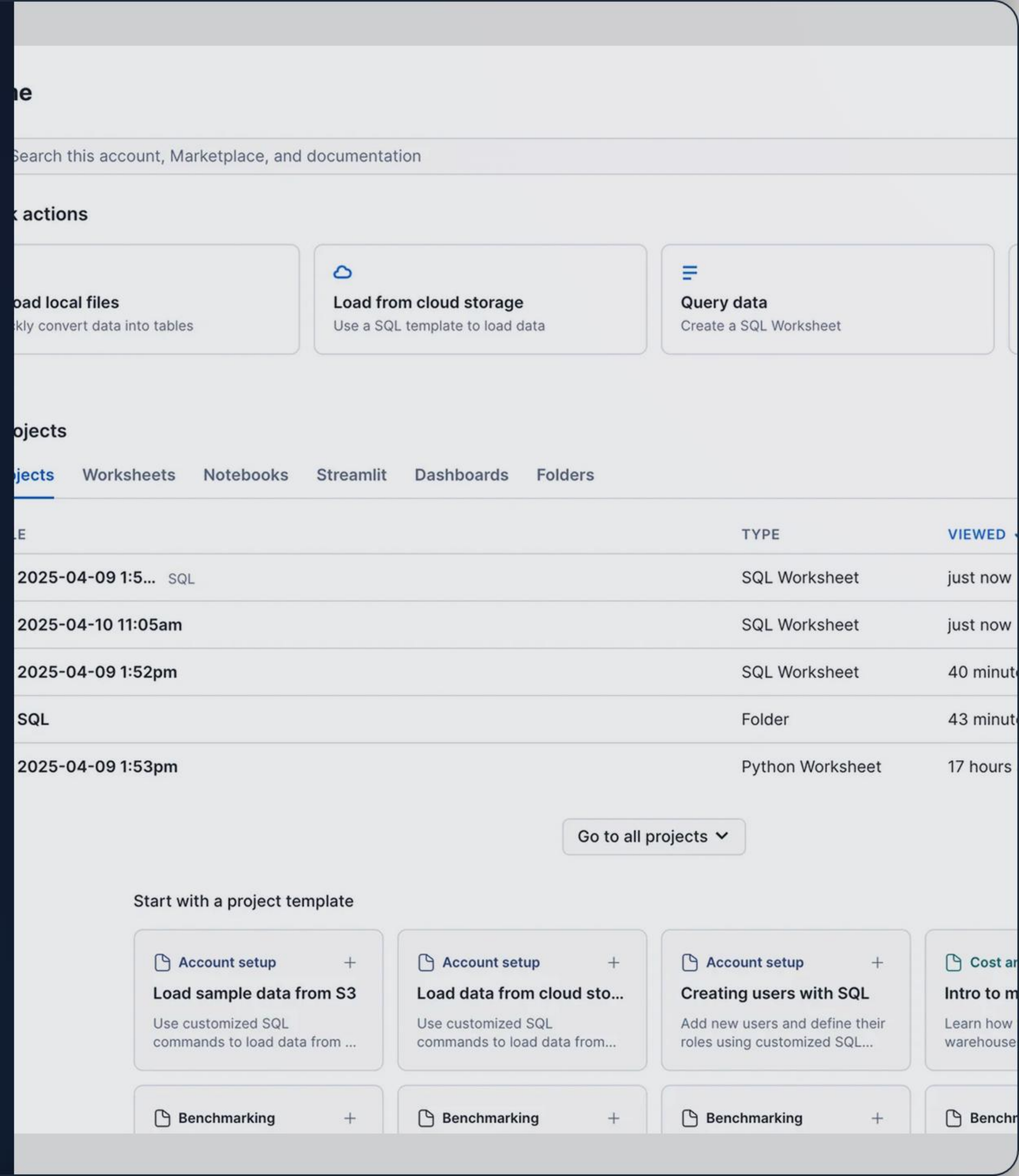
The WHERE Clause

WHERE = Condition. This is where the database becomes smart!



```
SELECT columns  
FROM table  
WHERE condition;
```

Filtering allows you to extract only the relevant rows that match your criteria, ignoring the rest of the dataset. It transforms raw data into actionable insights.



Filtering Examples



Numeric Filter

Students with marks above 80:

```
SELECT * FROM  
students  
WHERE marks > 80;
```



Exact Match

Only Class 5 students:

```
SELECT * FROM  
students  
WHERE class = '5th';
```



Text Search

Find a specific student:

```
SELECT * FROM  
students  
WHERE name = 'Rahul';
```


Comparison Operators

SYMBOL	MEANING	EXAMPLE QUERY USAGE
=	Equal	WHERE class = '5th'
>	Greater Than	WHERE marks > 80
<	Less Than	WHERE marks < 50
≥	Greater / Equal	WHERE marks ≥ 80
≤	Less / Equal	WHERE marks ≤ 75
≠	Not Equal	WHERE name ≠ 'Anita'

Real-Life Analogy

The Request (SELECT)

Imagine the **Database** is a giant Library.

Writing a **SELECT** query is just like walking up to the librarian and asking for books. You are initiating a search through a vast collection.

The Criteria (WHERE)

The **WHERE** clause is giving the specific condition.

Instead of just asking for "books", you say: *"Only give me Science Fiction books"*. The **Result** is getting the exact correct books instantly.

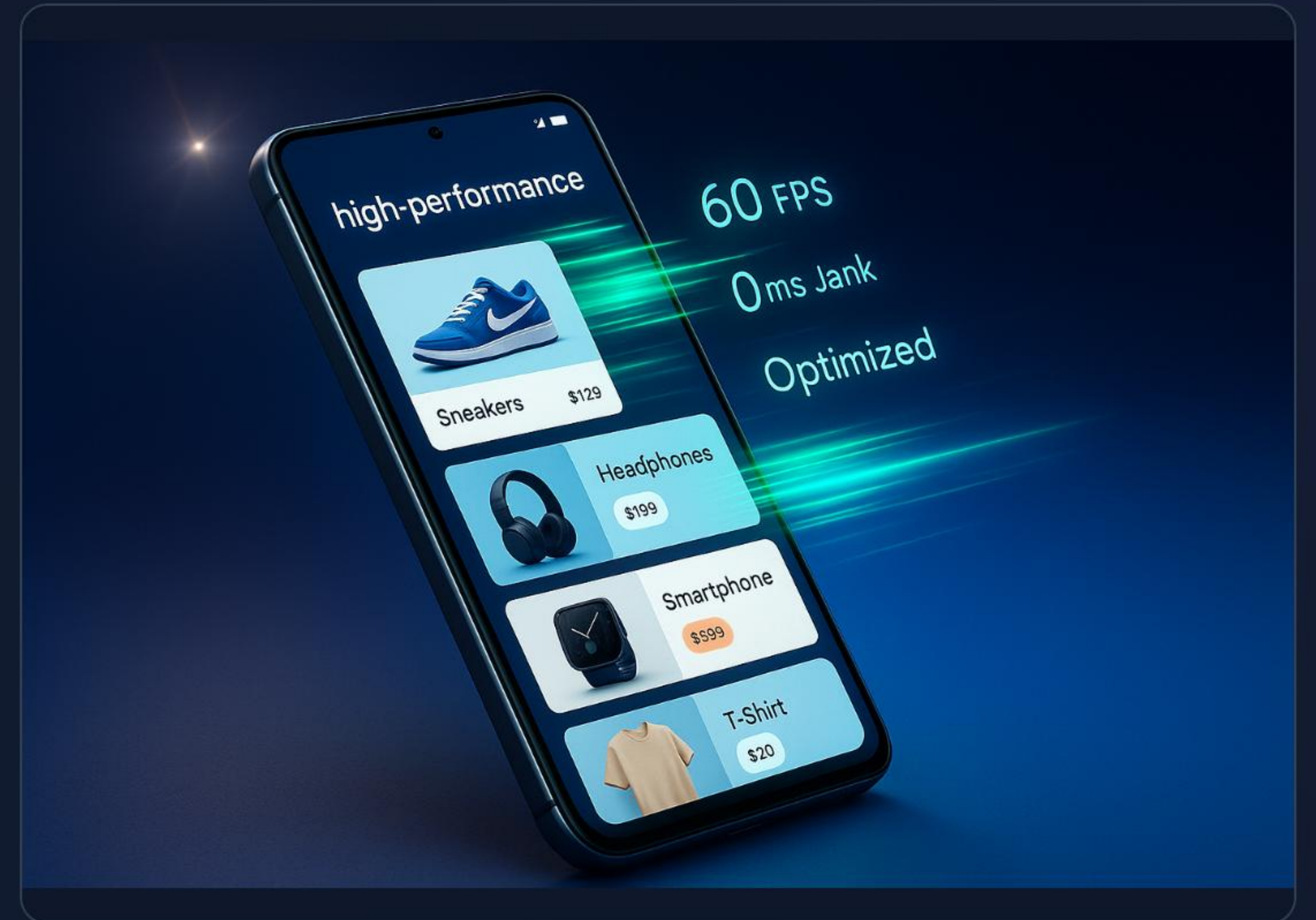
Mobile Shop Practice

Table Creation:

```
CREATE TABLE mobiles(  
  id INT,  
  brand VARCHAR(50),  
  price INT,  
  stock INT  
);
```

Practice Queries to Write:

-  Show all mobiles
-  Show brand and price
-  Mobiles with price > 20000
-  Mobiles where stock < 5



Thinking Challenge

”

Which query shows:
Students NOT in class 5th?

”

Answer:

```
SELECT * FROM students  
WHERE class ≠ '5th';
```


Bonus Skills

AND Condition

```
SELECT * FROM students  
WHERE class='5th' AND marks > 85;
```

When using **AND**, **BOTH** conditions must be TRUE. It makes your search much stricter and narrows down the results.

OR Condition

```
SELECT * FROM students  
WHERE class='5th' OR marks > 85;
```

When using **OR**, **ANY** condition can be TRUE. It broadens your search and brings back more results.

After Day 2 You Can...

-  **Search data like Google:** Look through massive datasets with a single line of text.
-  **Filter records instantly:** Pinpoint exact details without manual scrolling.
-  **Ask smart questions:** Communicate directly with the database engine.
-  **Think like an analyst:** Begin exploring data structures logically.



Tomorrow Preview: Day 3 🚀



Sorting

Learn to use **ORDER BY** to arrange your data logically (A-Z, High to Low).



Aggregation

Use **COUNT**, **SUM**, **AVG** to calculate crucial metrics from raw numbers.



Grouping

Master **GROUP BY** & **HAVING**.
This is where SQL becomes a real business tool!

Image Sources



<https://media.istockphoto.com/id/2197885290/photo/futuristic-flowing-data-stream.jpg?s=612x612&w=0&k=20&c=80G-rLZBCCZCyjawQOZ03oHjT0R4joisGr39DYOqnyI=>

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